Hexagonal Lattice Redemption Theory

Gauge Unification from Discrete Spacetime Geometry

HLRT-GUT: Deriving the Standard Model Gauge Structure and Coupling Hierarchy

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Abstract

We present Hexagonal Lattice Redemption Theory (HLRT) as a framework for gauge unification, deriving the Standard Model gauge groups and their coupling hierarchy from a single geometric principle: spacetime as a discrete hexagonal lattice with characteristic spacing $\lambda = 1.24 \times 10^{-13}$ m. The framework achieves unification by showing that U(1), SU(2), and SU(3) emerge naturally from distinct geometric structures on the lattice—hexagonal plaquettes, boundary-spanning transporters, and dual triangular plaquettes respectively.

The central result is a geometric derivation of the Standard Model gauge hierarchy. Through 7-cell hexagonal coarse-graining (real-space renormalization), we prove that the blocking factors satisfy $Z_1 < Z_2 < Z_3$, establishing $\alpha_1 < \alpha_2 < \alpha_3$ from lattice topology alone—without supersymmetry or extra dimensions. This spatial coarse-graining sets initial conditions compatible with standard momentum-space renormalization group flow.

HLRT predicts two gravitational propagation regimes: standard light-speed propagation in unperturbed spacetime, and modified dispersion within bounded fold regions at terahertz frequencies, permitting enhanced group velocities ($v_{GW} \approx 1.16c$) while signal velocity remains exactly c, preserving causality. The framework yields specific particle physics predictions including proton decay ($\tau_p \approx 2.5 \times 10^{35}$ years) and neutrino mass ($m_\nu \approx 0.050$ eV) derived from minimal lattice excitation.

Electromagnetic-lattice coupling predicts 5.5 ± 0.5% current enhancement in hexagonal configurations—a near-term experimental test. HLRT contains no adjustable free parameters; all constants are derived from lattice geometry or fixed by phenomenological matching. The theory is explicitly falsifiable. Extensions to dark sector physics, cosmology, and geophysics are briefly noted as directions for future work.

1. Introduction

The search for a unified framework connecting gravity with the Standard Model gauge forces has driven theoretical physics for a century. Grand unified theories (GUTs) typically postulate gauge symmetry at high energies, with symmetry breaking producing the observed structure. We present an alternative: Hexagonal Lattice Redemption Theory (HLRT), which *derives* the gauge groups from spacetime geometry rather than postulating them.

HLRT's central insight is that spacetime discretization at scale $\lambda \approx 1.24 \times 10^{-13}$ m provides a geometric substrate from which gauge dynamics emerge. The U(1), SU(2), and SU(3) gauge groups arise naturally from hexagonal plaquettes, boundary-spanning transporters, and dual triangular structures respectively. Most remarkably, the hierarchy of gauge couplings emerges from a coarse-graining theorem: why electromagnetism is weaker than the weak force, which is weaker than the strong force, follows from lattice topology.

This paper presents the gauge unification framework. Section 2 develops the gravitational sector and lattice foundation. Section 3 derives gauge field emergence and the coupling hierarchy, including the relationship to standard RG. Section 4 presents particle physics predictions. Section 5 specifies experimental tests and falsification criteria. Section 6 presents the unified action. Section 7 notes extensions to cosmology and geophysics deferred to future work.

2. Gravitational Sector and Lattice Foundation

2.1 Lattice Emergence

The hexagonal lattice emerges from scalar field self-interactions at the Planck scale. The fundamental lattice spacing derives from energy scale ratios:

$$\lambda = \mathrm{lP} \times (\mathrm{Efield} \,/\, \mathrm{EP})^{\mathrm{n}} \approx 1.24 \times 10^{-13}\ \mathrm{m}$$

where $\mathrm{lP} = 1.616 \times 10^{-35}$ m is the Planck length, $\mathrm{EP} = 1.956 \times 10^{9}$ J is the Planck energy, and n = 1. This scale—22 orders of magnitude above the Planck length—enables lattice effects to cascade to observable phenomena while maintaining discrete structure at sub-nuclear scales.

2.2 Modified Spacetime Metric

HLRT modifies Einstein's field equations through lattice field coupling:

$$\mathrm{ds}^2 = -\mathrm{f}(\mathrm{r},\Phi)\mathrm{c}^2\mathrm{dt}^2 + \mathrm{h}(\mathrm{r},\Phi)(\mathrm{dx}^2 + \mathrm{dy}^2 + \mathrm{dz}^2)$$

The lattice field $\Phi(r)$ couples to electromagnetic sources. Fold regions—where modified physics applies—are finite and mesoscopic ($\mathrm{Rfold} \lesssim 10^3$ km), explicitly excluding astrophysical extents. The metric functions incorporate geometric coupling Ageom = 0.0256 derived from hexagonal symmetry.

2.3 Two-Regime Gravitational Propagation

HLRT predicts two distinct propagation regimes:

- **Regime A (Standard):** Outside fold regions, vGW = c, consistent with GW170817 and all astrophysical constraints.
- **Regime B (Modified):** Inside bounded fold regions at THz frequencies with EM coupling, enhanced group velocity $\mathrm{vg} \approx 1.16\mathrm{c}$ along preferred lattice axes.

$$\mathrm{vGW}(\mathrm{r},\omega) = \mathrm{c} \times [1 + \beta\mathrm{fold} \times \Theta(\mathrm{Rfold} - \mathrm{r}) \times \mathrm{f}(\omega/\omega\mathrm{THz})]$$

where $\beta_{fold} = 0.16$ and $f(\omega/\omega_{THz}) = [1 + (\omega/\omega_{THz})^2]^{-1}$ ensures standard dispersion at high frequencies.

2.4 Signal Velocity and Causality

Theorem (Signal Velocity Bound): The front velocity v_f—determining information propagation—is bounded by c regardless of group velocity:

$$v_f = \lim_{\omega\to\infty} [\omega/k] = c$$

As $\omega \to \infty$, $f(\omega/\omega_{THz}) \to 0$, restoring standard dispersion. This parallels established anomalous dispersion physics (Wang et al. 2000, Stenner et al. 2003). Causality is preserved. □

2.5 Energy Conservation and CTC Prevention

The fold metric admits a timelike Killing vector, yielding conserved energy. Enhanced group velocity arises from local field energy redistribution, not creation.

Theorem (CTC Prevention): No closed timelike geodesic exists. For any closed curve γ, energy conservation requires $\oint_\gamma dt > 0$, while closure requires $\oint dt = 0$—a contradiction. □

3. Gauge Field Emergence from Lattice Geometry

This section presents the central contribution of HLRT-GUT: deriving the Standard Model gauge structure from lattice topology.

3.1 Lattice Gauge Theory Framework

We adopt the Wilson lattice gauge theory framework, where gauge fields live on lattice links as group-valued parallel transporters Uij ∈ G. The dynamics are governed by Wilson actions measuring deviation from trivial holonomy around closed loops. HLRT's contribution is showing that each Standard Model gauge group naturally inhabits a different geometric structure.

3.2 U(1) Electromagnetic Sector

Definition: The U(1) gauge field lives on edges of the hexagonal lattice. Link variables $U_{ij}^{(EM)} = \exp(ieA_{ij}) \in U(1)$. The elementary plaquette is the hexagonal face itself—a closed loop of 6 edges.

$$\mathrm{SU}(1) = -\beta_1\, \Sigma\mathrm{hex}\, [1 - \mathrm{Re}(\mathrm{Uhex})]$$

with $\beta_1 = 1/e^2$. The hexagonal plaquette—the *largest* elementary closed loop on the lattice—naturally hosts the *longest-range* force.

3.3 SU(2) Weak Sector

Definition: The SU(2) gauge field lives on edges with link variables $U_{ij}^{(weak)} \in SU(2)$. The relevant structures are *boundary-spanning transporters*: path-ordered products across one side of a super-hexagon boundary. Each side spans 3 fine-lattice edges.

$$\mathrm{SSU}(2) = -\beta_2\, \Sigma\mathrm{sides}\, [1 - ½\, \mathrm{Re\ Tr}(\mathrm{Uside})]$$

with $\beta_2 = 4/g_2^2$. This formulation directly identifies coarse link variables as products of fine link variables—a natural lattice construction.

3.4 SU(3) Strong Sector

The hexagonal lattice lacks intrinsic triangular plaquettes. However, its *dual lattice* is triangular.

Definition (Dual Lattice): The dual of hexagonal lattice H is triangular lattice T, whose vertices are hex centers and edges connect adjacent hex-centers. Elementary plaquettes of T are equilateral triangles.

Definition (SU(3) Sector): The SU(3) gauge field lives on edges of the dual triangular lattice T. Link variables $U_{ij}^{(color)} \in SU(3)$. The elementary plaquette is a triangle holonomy.

$$\mathrm{SSU}(3) = -\beta_3\, \Sigma\triangle\, [1 - ⅓\, \mathrm{Re\ Tr}(\mathrm{U}\triangle)]$$

with $\beta_3 = 6/g_3^2$. The 3 edges per triangle naturally reflect the triadic nature of color charge. These are the *smallest* plaquettes—most local, most robust to coarse-graining.

3.5 Geometric Unification: Why These Assignments?

The gauge group assignments are not arbitrary—they follow from the lattice's intrinsic geometry:

- **U(1):** Lives on the hexagonal lattice itself. Plaquettes are hexagons (6 edges). Largest elementary loops → longest-range force.

- **SU(2):** Lives on edges, with dynamics governed by boundary transporters (3 edges). Intermediate structure → intermediate-range force.
- **SU(3):** Lives on the dual triangular lattice. Plaquettes are triangles (3 edges). Smallest loops → shortest-range (confined) force.

This is the *unique* assignment respecting lattice topology. The hierarchy of spatial extent directly produces the hierarchy of forces.

3.6 The Coupling Hierarchy from Coarse-Graining

We now prove that the Standard Model gauge hierarchy emerges geometrically.

Definition (7-Cell Hexagonal Blocking): A blocking transformation replaces a cluster of 7 hexagons (one central plus six nearest neighbors) with a single effective "super-hexagon." This "flower" configuration has:

$$V = 24 \text{ vertices}, \quad E = 30 \text{ edges}, \quad F = 7 \text{ faces}$$

satisfying the planar Euler relation $V - E + F = 1$. Under blocking, linear scale increases as $a' \sim \sqrt{7}\, a$.

Lemma (Coarse-Graining Factors): Under 7-cell blocking, each gauge coupling rescales as $g'^2_i = Z_i\, g_i^2$, where Z_i is determined by the ratio of fine to coarse geometric structures.

U(1): 7 hexagonal plaquettes integrate out to 1 super-loop. $Z_1 = 1/7$.
SU(2): Each coarse boundary link is the product of 3 fine links. $Z_2 = 1/3$.
SU(3): Dual triangles are maximally local—minimal internal structure to integrate out. $Z_3 \approx 1$.

Theorem (Geometric Coupling Hierarchy): Under 7-cell hexagonal coarse-graining:

$$\mathbf{Z_1 = 1/7 < Z_2 = 1/3 < Z_3 \approx 1}$$

$$\mathbf{\Longrightarrow \quad \alpha_1 < \alpha_2 < \alpha_3}$$

The Standard Model gauge hierarchy—electromagnetism weaker than weak force weaker than strong force—emerges from hexagonal lattice topology. No supersymmetry, extra dimensions, or fine-tuning required. □

Summary of Gauge Emergence:

Group	Structure	Edges	Z	Range	Force
U(1)	Hex plaquettes	6	1/7	Long	EM
SU(2)	Boundary transp.	3	1/3	Medium	Weak
SU(3)	Dual triangles	3	≈ 1	Short	Strong

3.7 Relationship to Standard Renormalization Group

A natural question arises: standard RG analysis shows gauge couplings *running toward unification* at high energies. How does HLRT's coarse-graining relate?

The resolution distinguishes two types of renormalization:

- **Momentum-space RG:** Integrates out high-momentum modes. Couplings run with energy scale. Standard perturbative QFT.
- **Real-space RG (HLRT):** Integrates out short-distance degrees of freedom via spatial blocking. Couplings transform under decimation. Lattice/condensed matter approach.

These are *complementary, not contradictory*. HLRT's Z-factors establish the *initial hierarchy* at the lattice scale. Standard momentum-space RG then describes evolution from that initial condition. The ordering $\alpha_1 < \alpha_2 < \alpha_3$ is *preserved* under perturbative running because SM β-functions don't cause crossings at observable scales.

4. Particle Physics Predictions

4.1 Proton Decay

HLRT predicts proton decay through lattice-mediated baryon number violation:

$$\tau p = \hbar\lambda^2/(Gm^2p) \approx 2.5 \times 10^{35} \text{ years}$$

This satisfies current Super-Kamiokande constraints ($\tau p > 1.67 \times 10^{34}$ years) while remaining testable by Hyper-Kamiokande and DUNE.

4.2 Neutrino Mass

Neutrino mass derives from the minimal sustainable lattice excitation. The lattice energy scale is $\hbar c/\lambda \approx 994$ keV. With topologically-derived coupling $\alpha lattice \approx 10^{-6}$ (from hexagonal coordination number z = 3):

$$m\nu = (\hbar c/\lambda) \times \sqrt{(\alpha lattice/\alpha EM)} \approx 0.050 \text{ eV}$$

This is consistent with NOvA/KamLAND constraints ($\Sigma m\nu < 0.12$ eV). The neutrino represents the "quantum of lattice vibration"—the smallest mass-energy packet coherently traversing discrete spacetime.

4.3 Graviton Mass

Graviton propagation exhibits scale-dependent effective mass:

$$meff(E,\Phi) = m_0 \times [1 + \beta(E/E_0)^2 \times (\Phi/\Phi_0)^2]$$

where $m_0 \approx 10^{-22}$ eV/c² at cosmological scales. This effective mass lies below current observational bounds and does not modify astrophysical GW propagation in Regime A. The parameter $\beta \approx 0.1$ is fixed by matching to $\beta fold$.

5. Experimental Predictions and Falsification Criteria

5.1 Electromagnetic-Lattice Coupling

HLRT predicts measurable current enhancement in hexagonal geometries:

$$\Delta I/I_0 = 5.5 \pm 0.5\%$$

This enhancement should be symmetry-selective (present in hexagonal, absent in non-hexagonal controls), reproducible (>3σ across trials), and artifact-resistant. Thermal decoherence bound: Tmax $\lesssim$ 10 K. Room-temperature observation would imply collective enhancement factors.

5.2 Experimental Domain Clarification

HLRT's predictions are domain-specific. The lattice scale $\lambda \sim 10^{-13}$ m corresponds to ~1.6 GeV, within LHC reach. However, HLRT does *not* predict universal discreteness in standard scattering.

Why LHC constraints do not apply: LHC probes high-energy scattering in Regime A (unperturbed spacetime). Lorentz invariance is preserved there. HLRT's modified dispersion appears only in Regime B: bounded, mesoscopic regions with specific EM coupling at THz frequencies.

Correct experimental tests: (1) Geo-EM Amplifier for current enhancement; (2) High-frequency GW detectors (BREAD) for THz signatures; (3) Proton decay experiments; (4) Precision neutrino mass measurements.

5.3 Falsification Criteria

HLRT-GUT is explicitly falsifiable. The theory is disproven if:

1. EM enhancement consistently outside 5.0-6.0% range with >3σ significance under proper controls
2. Proton decay detected at rates inconsistent with $\tau p \approx 2.5 \times 10^{35}$ years
3. Neutrino mass measurements exclude the 0.048-0.053 eV range
4. Gauge hierarchy observed to violate $Z_1 < Z_2 < Z_3$ ordering under any conditions
5. Repeated targeted CMB template searches show no J_6 correlations at predicted scales

6. The Unified Action

The complete HLRT-GUT framework is described by:

$$\mathrm{SHLRT} = \mathrm{Sgravity} + \mathrm{SU(1)} + \mathrm{SSU(2)} + \mathrm{SSU(3)} + \mathrm{Smatter} + \mathrm{Slattice}$$

where:

- **Sgravity:** Einstein-Hilbert action with lattice field coupling
- **SU(1), SSU(2), SSU(3):** Wilson actions on respective geometric structures
- **Smatter:** Standard Model fermions with Higgs mechanism
- **Slattice:** Lattice field dynamics including nonlinear self-interaction

All gauge coupling ratios are determined by geometric Z-factors. No parameters are adjustable post hoc.

7. Extensions and Future Work

HLRT-GUT forms the foundation for broader extensions currently under development:

- **Dark Sector:** Dark matter as topological lattice defects (disclinations); dark energy from lattice vacuum energy. Preliminary formalism exists but requires N-body simulation.
- **Cosmology:** CMB hexagonal correlations at arcminute scales; primordial perturbation seeding of defect networks.
- **Geophysics (CDGR):** Core Displacement and Geodynamic Rebalancing extending lattice dynamics to Earth's interior. Pole drift relaxation model with $\tau \approx 25$-40 years, reflecting uncertainty in core-mantle coupling.
- **Quantum Gravity:** Full treatment of how QM and GR both emerge from lattice dynamics; information paradox resolution via discrete structure.

These extensions will be presented in a companion Theory of Everything paper (HLRT-ToE) following experimental validation of core predictions.

8. Conclusions

Hexagonal Lattice Redemption Theory provides a framework for gauge unification grounded in discrete spacetime geometry. The central achievements are:

- **Gauge emergence:** U(1), SU(2), SU(3) derived from hexagonal plaquettes, boundary transporters, and dual triangles

- **Hierarchy theorem:** $Z_1 < Z_2 < Z_3$ from 7-cell coarse-graining, compatible with standard RG
- **Particle predictions:** Proton decay and neutrino mass from lattice physics
- **Near-term test:** 5.5% EM enhancement in hexagonal configurations

The framework operates in experimentally unexplored regimes and does not contradict existing constraints. All predictions are quantitative and falsifiable. If validated, HLRT-GUT provides a geometric foundation for the Standard Model. The experiment decides.

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Appendix: Parameter Summary

All parameters are derived from lattice geometry or fixed by phenomenological matching. None are adjustable post hoc.

Parameter	Symbol	Value	Origin
Lattice Spacing	λ	1.24×10^{-13} m	Derived
U(1) Z-factor	Z_1	1/7	Geometric
SU(2) Z-factor	Z_2	1/3	Geometric
SU(3) Z-factor	Z_3	≈ 1	Geometric
Hexagonal Anisotropy	β_{fold}	0.16	Geometric
Lattice Coupling	$\alpha_{lattice}$	$\approx 10^{-6}$	Topological
EM Enhancement	$\Delta I/I_0$	$5.5 \pm 0.5\%$	Derived
Proton Decay	τ_p	2.5×10^{35} yr	Derived
Neutrino Mass	m_ν	0.050 eV	Derived
Graviton Mass (cosmo)	m_0	$\approx 10^{-22}$ eV	Below bounds

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